

FULLERTON, CA, USA

WMO: 722976

Lat: 33.872N

Lon: 117.979W

Elev: 29

StdP: 100.97

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 03166

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	4.5	6.0	-12.9	1.2	20.0	-9.3	1.7	21.0	8.5	19.3	7.3	18.1	0.5	90	0.327

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.6	35.2	19.3	33.2	19.3	31.5	19.0	22.4	30.8	21.7	29.6	21.0	28.4	2.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	14.5	25.8	18.9	13.8	25.2	18.1	13.1	24.2	66.0	30.9	63.2	29.5	60.9	28.6	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
5.7	4.9	4.5	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			1.8	39.9	1.3	2.4	0.8	41.6	0.1	43.0	-0.7	44.4	-1.6	46.1		
			-0.3	23.5	1.2	1.0	-1.2	24.2	-1.8	24.8	-2.5	25.3	-3.4	26.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.9	14.4	14.6	16.2	17.6	19.3	21.4	23.9	24.4	23.6	20.7	17.1	13.8
	DBStd	4.55	2.86	2.90	2.86	2.91	2.45	2.14	2.18	2.28	2.91	2.86	2.86	2.48
	HDD10.0	5	2	1	0	0	0	0	0	0	0	0	0	2
	HDD18.3	579	125	109	77	47	15	1	0	0	0	8	56	141
	CDD10.0	3269	139	129	193	227	289	341	432	447	407	331	213	121
	CDD18.3	802	4	4	12	24	46	92	173	189	157	81	18	1
	CDH23.3	6324	118	107	178	278	314	502	1184	1411	1217	707	258	51
	CDH26.7	2402	25	32	57	108	120	143	417	562	533	307	92	6
Wind	WSAvg	1.6	1.4	1.6	1.7	2.0	2.0	2.0	1.8	1.7	1.5	1.4	1.3	1.3
Precipitation	PrecAvg	337	77	82	46	25	8	2	1	0	5	12	28	51
	PrecMax	670	310	346	158	63	61	29	10	3	42	133	79	249
	PrecMin	110	3	0	0	4	0	0	0	0	0	0	1	0
	PrecStd	157	81	76	44	14	13	7	2	1	9	26	20	53
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.4	30.7	32.2	34.5	35.2	34.1	34.9	36.1	38.3	36.7	33.1	27.4
		MCWB	13.1	13.9	15.5	15.3	15.7	21.2	21.4	21.2	21.0	17.2	15.2	12.9
	2%	DB	26.9	26.8	27.7	29.6	29.4	30.1	32.4	33.8	34.7	33.0	29.3	24.7
		MCWB	12.9	13.3	14.3	14.9	16.6	19.8	20.7	21.0	20.9	16.9	14.2	12.3
	5%	DB	24.0	23.4	24.8	26.3	26.6	27.8	30.7	32.0	32.2	30.1	26.2	22.3
		MCWB	12.1	12.8	13.7	14.7	17.3	19.2	20.6	20.7	20.1	16.3	13.5	11.7
	10%	DB	21.3	20.9	22.2	23.5	24.0	26.2	28.9	29.8	29.6	27.2	23.3	19.7
		MCWB	11.9	12.4	13.9	14.7	16.6	18.6	20.3	20.3	19.6	16.9	13.4	11.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.3	16.7	18.0	18.2	20.1	21.6	23.2	23.5	23.5	21.8	17.7	16.5
		MCDB	22.1	23.3	25.7	26.9	28.1	30.9	31.2	32.5	33.2	28.5	24.1	20.4
	2%	WB	15.1	15.4	16.5	16.9	18.7	20.5	22.1	22.3	22.2	20.1	16.8	15.3
		MCDB	19.5	21.0	23.2	25.1	25.9	28.5	29.5	30.8	31.0	27.3	23.1	19.0
	5%	WB	14.3	14.5	15.6	16.0	17.8	19.7	21.4	21.6	21.3	19.0	15.9	14.4
		MCDB	18.9	19.9	21.6	23.5	24.5	26.9	28.4	29.6	29.2	25.5	21.8	18.1
	10%	WB	13.5	13.7	14.8	15.2	16.9	18.9	20.7	20.8	20.5	18.0	15.2	13.5
		MCDB	18.2	18.6	20.0	21.4	22.7	25.3	27.4	28.1	27.7	23.8	20.5	17.3
Mean Daily Temperature Range	5% DB	MDBR	12.2	11.2	10.6	10.4	9.0	8.9	9.9	10.6	11.0	11.5	12.3	11.9
		MCDBR	17.8	16.8	16.3	16.7	14.5	12.5	12.5	13.9	15.2	18.0	17.8	16.9
		MCWBR	7.5	7.1	6.7	6.5	5.5	4.6	4.3	4.8	5.1	6.4	6.9	7.8
	5% WB	MCDBR	11.7	12.3	11.6	12.4	10.8	11.1	11.0	11.9	12.7	12.2	12.4	10.9
MCWBR		6.0	6.0	5.5	5.5	4.6	4.4	4.1	4.5	4.8	5.3	5.9	6.2	
Clear-Sky Solar Irradiance	taub	0.332	0.342	0.368	0.371	0.370	0.386	0.389	0.373	0.359	0.363	0.317	0.316	
	taud	2.510	2.492	2.393	2.391	2.428	2.384	2.444	2.475	2.505	2.465	2.593	2.594	
	Ebn at Noon	880	908	907	916	915	895	891	903	904	871	894	881	
	Edh at Noon	85	95	113	119	116	121	113	108	100	97	77	74	
All-Sky Solar Radiation	RadAvg	3.19	4.08	5.46	6.65	7.18	7.56	7.49	7.12	6.02	4.58	3.54	2.89	
	RadStd	0.22	0.40	0.29	0.33	0.46	0.45	0.39	0.21	0.23	0.32	0.20	0.21	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+1.16	N/A	-2.68	+1.32	N/A	N/A	N/A	-192	+489
(47) Regional (54 neighbors)		+0.62	N/A	-1.65	+0.70	N/A	N/A	N/A	-112	+220

Nomenclature: See separate page